

AI Image & Video Generation

Purpose:

This document explains what AI Image Generation and AI Video Generation are, how companies use them, the difference between using existing AI APIs and building your own AI service, estimated costs, and the recommended implementation process.

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1. Introduction

Artificial Intelligence (AI) is becoming an important technology in modern software development. Today, AI can generate text, images, videos, music, code, and many other types of digital content.

Instead of manually designing images or editing videos, users can simply type a description, and AI automatically generates the required output.

Many companies are integrating AI into their applications to improve productivity and provide advanced features to customers.

2. What is Artificial Intelligence (AI)?

Artificial Intelligence is a technology that enables computers to perform tasks that normally require human intelligence.

Examples include:

- Answering questions
- Writing content
- Translating languages
- Creating images
- Creating videos
- Understanding speech
- Recognizing objects

Examples of AI applications:

- ChatGPT
- Google Gemini
- Microsoft Copilot
- Claude

3. What is Generative AI?

Generative AI is a type of Artificial Intelligence that creates completely new content instead of simply searching for existing information.

It can generate:

- Images
- Videos
- Audio
- Documents
- Software Code
- Presentations

Example:

Prompt:

"A beautiful waterfall surrounded by mountains during sunset."

The AI generates a completely new image based on the description.

4. What is Image Generation AI?

Image Generation AI creates new images using text prompts.

Example:

Prompt:

"A modern university campus with students walking during sunset."

Output:

A completely new AI-generated image.

Popular Image AI Models:

- OpenAI GPT Image
- Stable Diffusion
- FLUX
- Google Imagen
- Midjourney

5. What is Video Generation AI?

Video Generation AI creates videos using:

- Text
- Images
- Existing Videos

Example:

Prompt:

"A lion walking through a forest."

Output:

A realistic video showing a lion walking in the forest.

Popular Video AI Models:

- OpenAI Sora

- Google Veo
- Runway
- Pika
- Kling AI

6. How Companies Use AI

Companies integrate AI into many business applications.

Examples:

Marketing

Generate posters

Generate advertisements

Generate social media banners

Education

Generate educational illustrations

Generate learning videos

Healthcare

Medical image analysis

Educational medical diagrams

E-commerce

Generate product images

Generate promotional banners

Software Companies

AI Chatbots

AI Image Generator

AI Video Generator

7. Two Methods to Implement AI

There are two common approaches.

Option 1

Using Existing AI APIs

Option 2

Building or Hosting Your Own AI

8. Option 1 – Using Existing AI APIs

In this method, your application communicates with an AI provider through an API.

Example providers:

- OpenAI
- Google
- Stability AI
- Fal.ai
- Replicate

Architecture:

User



Your Application



AI Provider API



Generated Image

↓

User

The AI provider manages:

- GPU
- AI Model
- Infrastructure
- Scaling
- Maintenance

9. Benefits of Existing AI APIs

Advantages:

- ✓ Fast implementation
- ✓ No GPU required
- ✓ High quality AI models
- ✓ Easy integration
- ✓ Automatic updates
- ✓ Suitable for startups
- ✓ Low initial investment

10. Limitations of Existing AI APIs

Disadvantages:

API usage charges

Internet connection required

Dependent on external provider

Monthly cost increases with higher usage

11. Step-by-Step Process

Step 1

Create an account.

Example:

OpenAI Platform

Create your account.

Step 2

Configure billing.

Add:

Credit Card

or

Supported Payment Method

Step 3

Create an API Key.

Example:

API Keys

↓

Create Secret Key

The generated key looks similar to:

sk-xxxxxxxxxxxxxxxxxx

Store it securely.

Never expose it publicly.

Step 4

Install the required SDK.

Example:

```
pip install openai
```

Step 5

Create your backend API.

Example:

POST

```
/api/generate-image/
```

Step 6

User enters:

```
Generate a tiger walking in snow.
```

Step 7

Your backend sends the request to the AI provider.

Step 8

AI generates the image.

Step 9

Save the image.

Possible storage:

AWS S3

Cloud Storage

Local Storage (Development)

Step 10

Return the image URL.

Frontend displays the image.

12. Approximate API Cost

Pricing depends on:

Provider

Image Resolution

Image Quality

Model Used

As a planning estimate:

Low Quality

Approximately ₹1 per image

Medium Quality

Approximately ₹3–₹5 per image

High Quality

Approximately ₹14–₹15 per image

These values are approximate and may change based on the provider's pricing.

Example:

100 images

≈ ₹400

1000 images

≈ ₹4,000

10,000 images

≈ ₹40,000

13. Option 2 – Self-Hosted AI

Instead of using another company's API, you host the AI model yourself.

Architecture:

User



Your Application



Your AI Server



GPU



AI Model



Generated Image

You own:

The API

The AI Model

The Server

The Database

The Storage

14. Benefits of Self-Hosted AI

No per-image API charges

Complete control

Data privacy

Model customization

Can reduce costs for very large workloads

15. Challenges

Requires GPU servers

Higher initial investment

More maintenance

Need AI deployment knowledge

Need DevOps experience

Need scaling infrastructure

16. Infrastructure Required

Typical software stack:

Ubuntu Linux

Docker

CUDA

PyTorch

AI Model

FastAPI

Django

React

Hardware:

GPU

CPU

RAM

SSD

Internet

Storage

17. Estimated Infrastructure Cost

Personal GPU Workstation

GPU

₹1.5 – ₹2 lakh

Complete System

₹2.5 – ₹4 lakh

Cloud GPU

Monthly cost depends on:

GPU model

Usage hours

Cloud provider

Typical range:

A few thousand rupees per month for light workloads to more than ₹2 lakh/month for continuously running high-end GPU instances.

18. Existing API vs Self-Hosted AI

Feature	Existing API	Self-Hosted AI
Initial Cost	Low	High
Development Time	Fast	Longer
Infrastructure	Managed by Provider	Managed by Company
GPU Required	No	Yes
Maintenance	Very Low	High
Per-Image Cost	Yes	No third-party API charge
Scalability	Easy	Company Responsibility
Best For	MVPs & Startups	Large-scale Products

20. Conclusion

For organizations beginning their AI journey, **using an existing AI API is the recommended approach**. It requires minimal infrastructure, allows faster development, and enables teams to validate their product quickly.

As the application grows and API usage increases, organizations can evaluate **hosting open-source AI models on their own GPU infrastructure** to gain greater control and potentially reduce long-term operational costs.